2 Program Logic: Selection, Basic Calculations and Operators

READ ME!

1. In all exercises in this exercise set, you should ***print the last line*** in the program's output using `System.out.`**`println`**. That is, there is a line break at the end of the output.
2. A fast way to write `System.out.println();` is as below:
   1. Type **syso**
   2. Press **Ctrl + Spacebar**
3. **READ THIS:** 👉 Keep pressing **Ctrl + Shift + F** to format your source text properly! 👈

Meal Program

Create a program called **MealProgram** that calculates the cost of a meal at a restaurant. The program should input the cost of the food, the tax rate, and the tip percentage (before tax) from the user. Then the program should calculate and print the total cost of the meal as shown in the example output.

- The program should accept both comma and dot as decimal part separator.

Example output:

```
Enter the price of food: 20.00
Enter the tax rate: 10.0
Enter the tip percentage: 15.0

The total cost is 25,00
```

Population Program

Create a program called **PopulationProgram** that first inputs the number of inhabitants of three areas from the user. Then the program should calculate the *percentage of population in each area*. The program should show the percentages with one decimal place as shown in the example output.

Example output:

```
Enter the population of the area A: 12139
Enter the population of the area B: 18764
Enter the population of the area C: 4313

A: 34,5 %
B: 53,3 %
C: 12,2 %
```

👉 Keep pressing **Ctrl + Shift + F** to format your source text properly!

Division and Remainder

Create a program called ***DivisionAndRemainder*** that first inputs two integers x and y from the user. Then the program should determine if the integers are even or odd. In addition, the program should compute the remainder of x / y and the remainder of y / x.

- You can suppose that the user does not ever enter a zero.
- Hint: See the ***arithmetic operators*** in the course material.

Example output:

```
Enter x: 12
Enter y: 9

12 is even
9 is odd
The remainder of 12 / 9 is 3
The remainder of 9 / 12 is 9
```

Soft Drinks

Create a program called ***SoftDrinks*** that first inputs the number of students and the number of soft drink bottles from the user. Then the program should compute how many bottles of soft drinks there are per student and how many leftover bottles there will be. Each student should get the same number of bottles.

- You can suppose that the user does not ever enter a zero.
- **NB!** You should use an ***arithmetic operator*** to determine the number of leftover bottles.

Example output:

```
Enter the number of students: 2
Enter the number of bottles: 6

There will be 3 bottles for each student.
There will be 0 leftover bottles.
```

Train Travel

Create a program called ***TrainTravel*** that first inputs the number of train trips the user makes per month, the price of a single train ticket, and the price of the monthly season ticket (valid for one month). Then the program should determine which one is the cheaper option (buying a monthly season ticket or buying single tickets) and how much cheaper it is.

Example output:

```
Enter the number of train trips: 3
Enter the single ticket price: 20,50
Enter the monthly season ticket price: 80,00

Buying single tickets is 18,50 euros cheaper.
```

The program should print one of the following texts:

Buying a monthly season ticket is 99,00 euros cheaper.

Buying single tickets is 99,00 euros cheaper.

The total cost is the same.

Beach Volleyball

Create a program called ***BeachVolleyball*** for dividing expenses in a group of people who rent a beach volleyball court.

The program should first input the number of people, the *hourly rate* of a court, and the length of rental in full hours. Then the program should calculate how much each of them should pay. The program should display the sum with two decimals.

- You can suppose that the user does not ever enter a zero.

Example output:

```
Enter the number of people: 5
Enter the hourly rate: 20,50
Enter the length of the rental (hours): 3

Each of the 5 people should pay 12,30 euros.
```

Multiplication Program

Create a program called **Multiplication program** that first inputs two decimal numbers from the user. Then the program should compute and print the product of the decimal numbers as shown in the example output.

- **NB!** The program should be to perform 100 % accurate computations on decimal values. It should always print a mathematically correct result.
- Please study the course materials carefully and see how we can perform ***accurate computations*** on decimal values.

Example output:

```
Enter x: 0.1
Enter y: 2.2

0.1 * 2.2 = 0.22
```

Example output:

```
Enter x: 2.345
Enter y: 3.141592653589238

2.345 * 3.141592653589238 = 7.367034772666763110
```